```
#include <Wire.h>
#include "rgb_lcd.h"

rgb_lcd lcd;

const int button = 2;
const long dotThreshold = 300;
const long letterThreshold = 800;
const long wordThreshold = 1250;

int lastButtonState = HIGH;
unsigned long pressedTime = 0;
unsigned long releasedTime = 0;

String currentMorse = "";
bool wordSpacePrinted = true; // Prevents endless spaces when idle

void translateMorse(String phrase) {
  if (phrase == ".....") {
    lcd.clear();
    lcd.setCursor(0,0);
  }
  else if (phrase == ".-") { lcd.print("A"); }
  else if (phrase == "-...") { lcd.print("B"); }
  else if (phrase == "-.-.") { lcd.print("C"); }
  else if (phrase == "-..") { lcd.print("D"); }
  else if (phrase == ".") { lcd.print("E"); }
  else if (phrase == "..-.") { lcd.print("F"); }
  else if (phrase == "--.") { lcd.print("G"); }
  else if (phrase == "....") { lcd.print("H"); }
  else if (phrase == "..") { lcd.print("I"); }
  else if (phrase == ".---") { lcd.print("J"); }
  else if (phrase == "-.-") { lcd.print("K"); }
  else if (phrase == ".-..") { lcd.print("L"); }
  else if (phrase == "--") { lcd.print("M"); }
  else if (phrase == "-.") { lcd.print("N"); }
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  else if (phrase == "---") { lcd.print("O"); }
  else if (phrase == ".--.") { lcd.print("P"); }
  else if (phrase == "--.-") { lcd.print("Q"); }
  else if (phrase == ".-.") { lcd.print("R"); }
  else if (phrase == "...") { lcd.print("S"); }
  else if (phrase == "-") { lcd.print("T"); }
  else if (phrase == "..-") { lcd.print("U"); }
  else if (phrase == "...-") { lcd.print("V"); }
  else if (phrase == ".--") { lcd.print("W"); }
  else if (phrase == "-..-") { lcd.print("X"); }
  else if (phrase == "-.--") { lcd.print("Y"); }
  else if (phrase == "--..") { lcd.print("Z"); }
  else { lcd.print("?"); }
}

void setup() {
    Serial.begin(9600);
    pinMode(button, INPUT_PULLUP);
    lcd.begin(16,2);
}

void loop() {
  int currentState = digitalRead(button);
  unsigned long currentTime = millis();

  // Button pressed
  if (lastButtonState == HIGH && currentState == LOW) {
    pressedTime = currentTime;
    delay(50);
  }

  // Button released
  else if (lastButtonState == LOW && currentState == HIGH) {
    unsigned long pressDuration = currentTime - pressedTime;
    releasedTime = currentTime;
    wordSpacePrinted = false; // Reset the flag so a new space can be printed later

    if (pressDuration > 20 && pressDuration <= dotThreshold) {
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      currentMorse += ".";
    }
    else if (pressDuration > dotThreshold) {
      currentMorse += "-";
    }

    delay(50);
  }

  // Handle pauses while the button is unpressed
  if (currentState == HIGH) {

    // 1. Check for letter completion
    if (currentMorse.length() > 0 && (currentTime - releasedTime > letterThreshold)) {
      translateMorse(currentMorse);
      currentMorse = "";
    }

    // 2. Check for word completion (only if we just finished a letter and haven't printed a space yet)
    else if (currentMorse.length() == 0 && !wordSpacePrinted && (currentTime - releasedTime >
wordThreshold)) {
      lcd.print(" ");
      wordSpacePrinted = true; // Lock the flag until the next button press
    }
  }

  lastButtonState = currentState;
}
```